

It's hard to understand something when you can't see it and don't know it exists! In this chapter, we'll pull back the curtain to dispel the overblown notion that recursion is hard, and you'll be able to appreciate the elegance underneath.

The Definition of Recursion

Before we discuss recursion, let's get the clichéd recursion jokes out of the way, starting with this: "To understand recursion, you must first understand recursion."

During the months I've spent writing this book, I can assure you that this joke gets funnier the more you hear it.

Another joke is that if you search Google for *recursion*, the results page asks if you mean *recursion*. Following the link, as shown in Figure 1-1, takes you to . . . the search results for *recursion*.

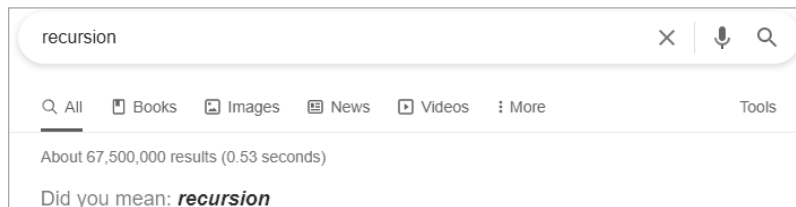


Figure 1-1: The Google search results for recursion link to the Google search results for recursion.

Figure 1-2 shows a recursion joke from the webcomic xkcd.



Figure 1-2: I'm So Meta, Even This Acronym (I.S. M.E.T.A.)

Most jokes about the 2010 science-fiction action movie *Inception* are recursion jokes. The film features characters having dreams within dreams within dreams.

And finally, what computer scientist could forget that monster from Greek mythology, the recursive centaur? As you can see in Figure 1-3, it is half horse, half recursive centaur.

